

SUPPLEMENTARY MATERIAL FOR “TORUS COMPUTED TOMOGRAPHY FOR EXPERIMENTAL DATA”

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Supplementary material to the article “Torus computed tomography for experimental data” (arXiv:2608.16453).

This supplement contains the regularization error surfaces for the simulated data experiments of Section 4 of the main paper. Notation and error measures are as in the main paper.

REGULARIZATION ERROR SURFACES

The regularization parameters α and s were selected by minimizing the reconstruction error over a two-dimensional grid for each phantom separately. The search was performed over $\alpha \in [0, 10^{-4}]$ (21 uniformly spaced values) and $s \in [0, 2]$ (41 uniformly spaced values). Figures S1–S9 show the L^1 , L^2 and L^∞ errors as functions of α and s for all three phantoms, without (left column) and with (right column) the positivity constraint. The white cross indicates the minimizer of the corresponding error matrix.

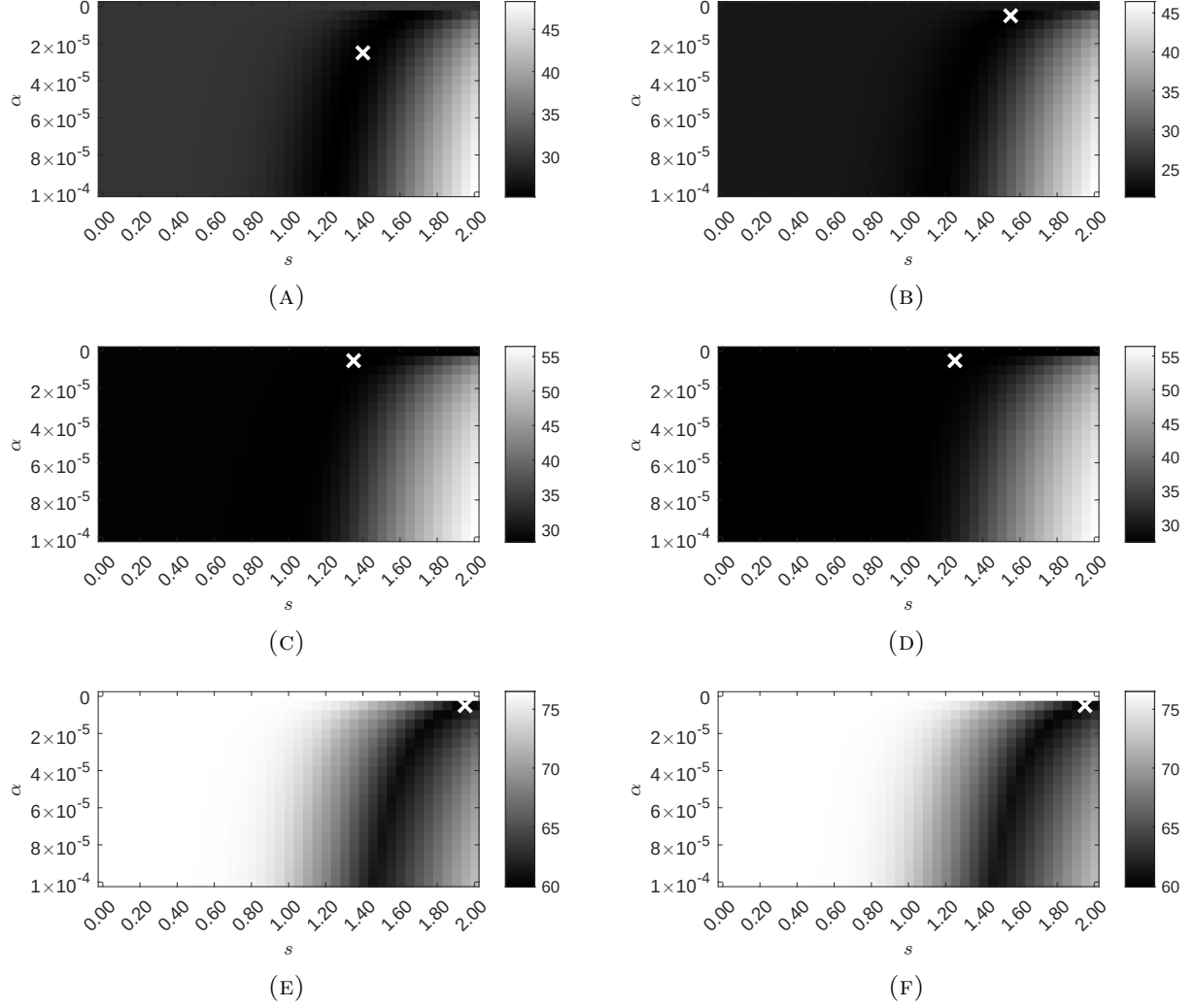


FIGURE S1. Torus CT reconstruction errors as functions of the regularization parameters α and s for the Shepp–Logan phantom, without (left column) and with (right column) the positivity constraint. Rows correspond to the L^1 , L^2 , and L^∞ error metrics.

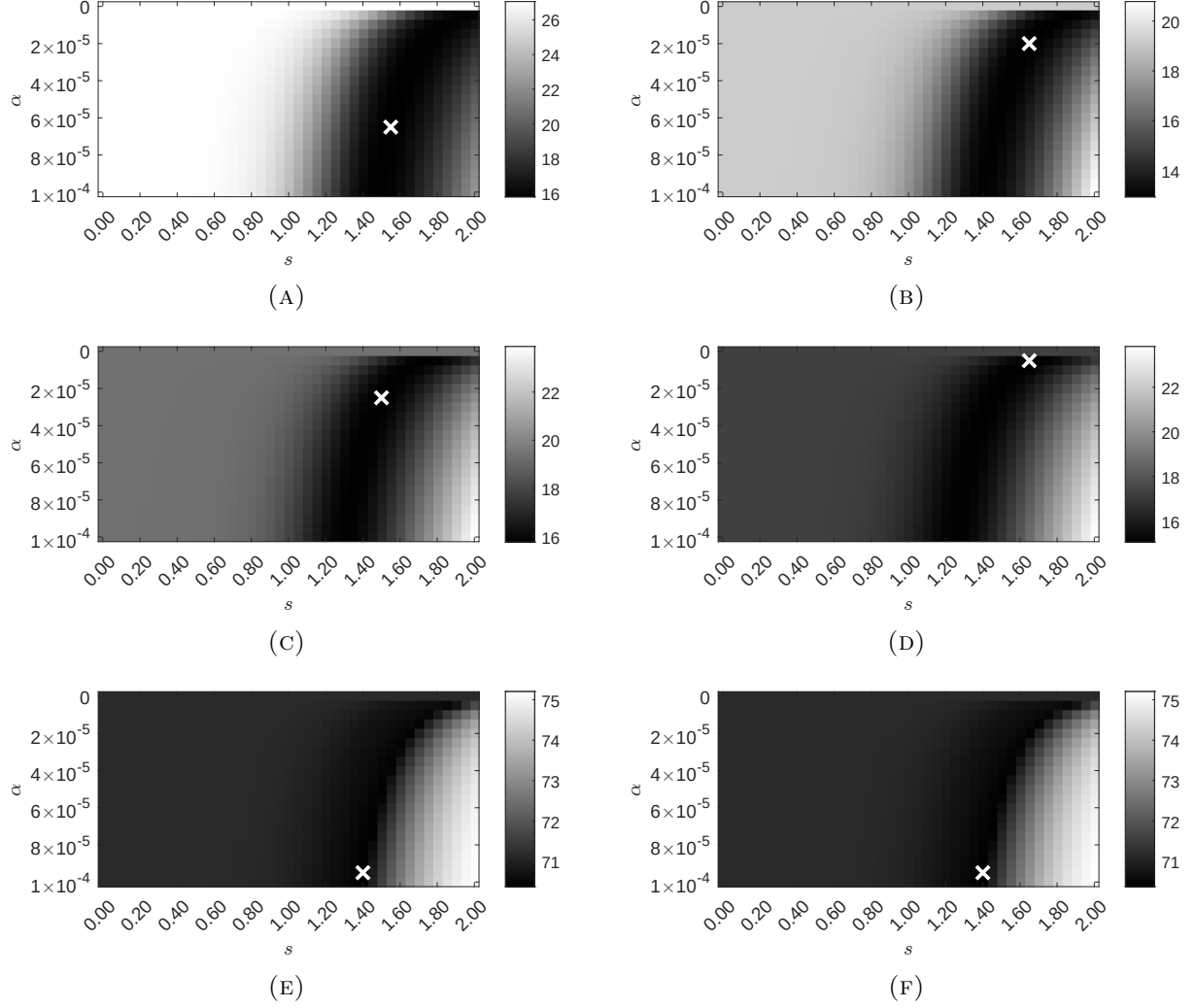


FIGURE S2. Torus CT reconstruction errors as functions of the regularization parameters α and s for the flag phantom, without (left column) and with (right column) the positivity constraint. Rows correspond to the L^1 , L^2 , and L^∞ error metrics.

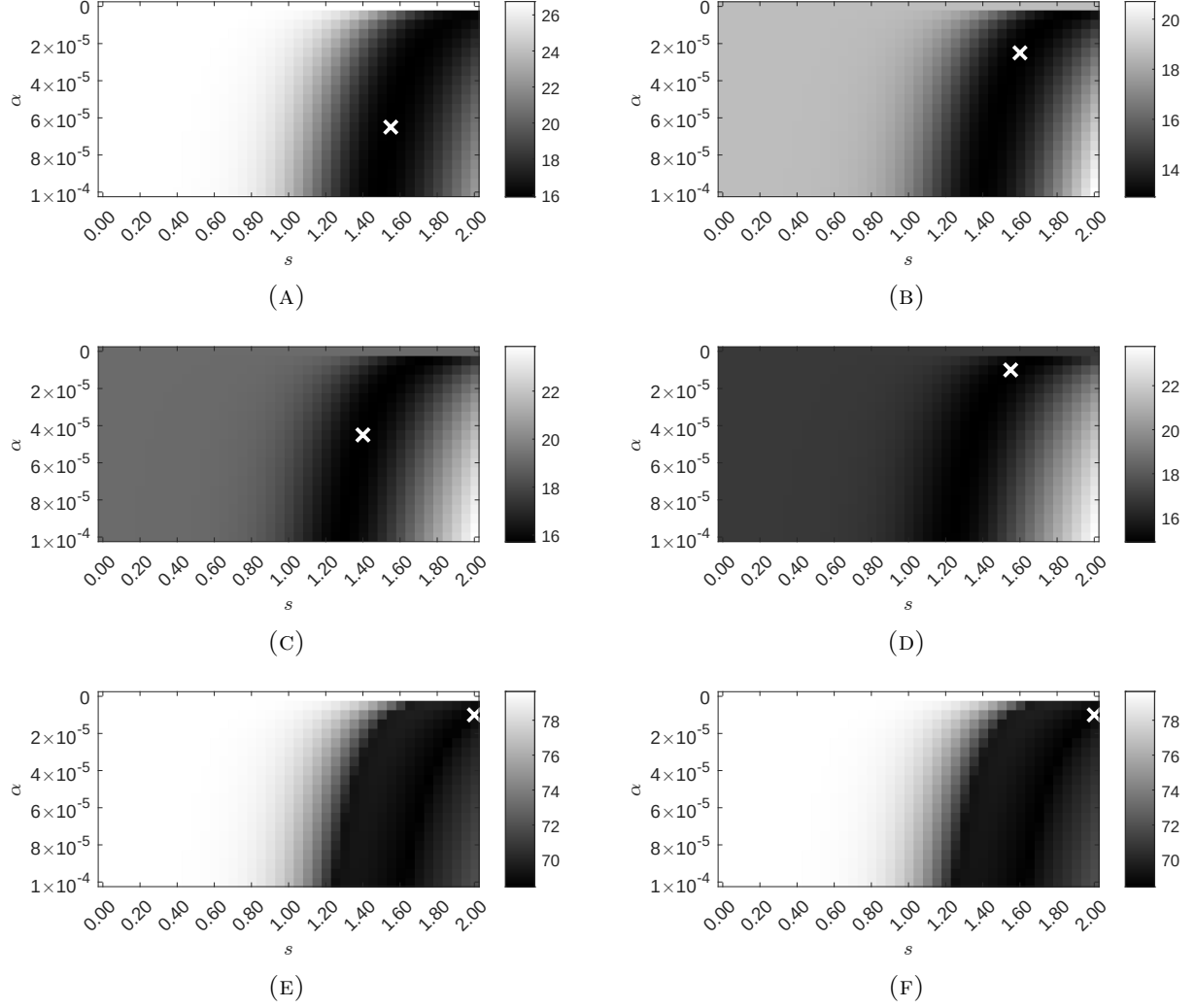


FIGURE S3. Torus CT reconstruction errors as functions of the regularization parameters α and s for the rotated flag phantom, without (left column) and with (right column) the positivity constraint. Rows correspond to the L^1 , L^2 , and L^∞ error metrics.

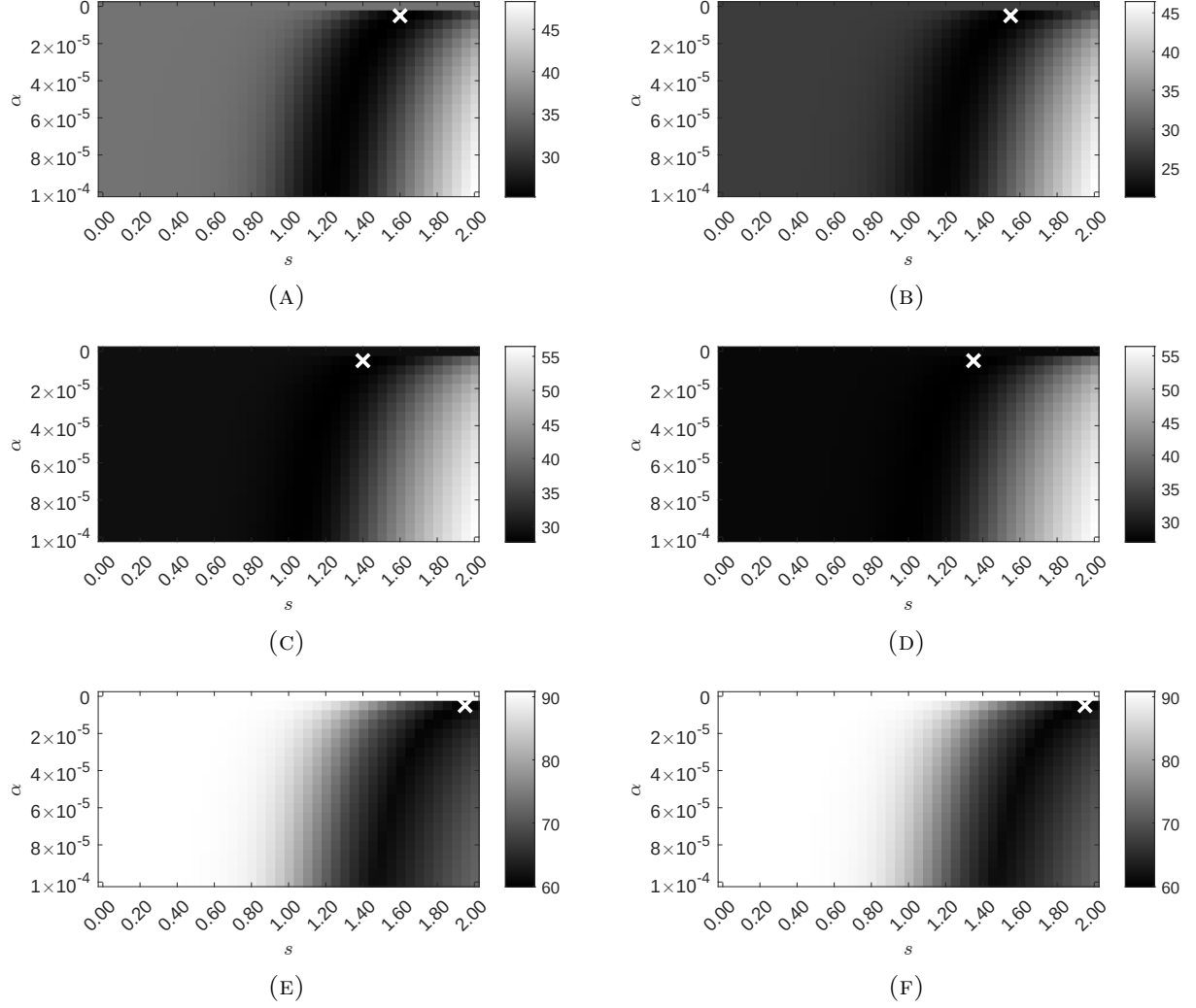


FIGURE S4. Star TCT reconstruction errors as functions of the regularization parameters α and s for the Shepp–Logan phantom, without (left column) and with (right column) the positivity constraint. Rows correspond to the L^1 , L^2 , and L^∞ error metrics.

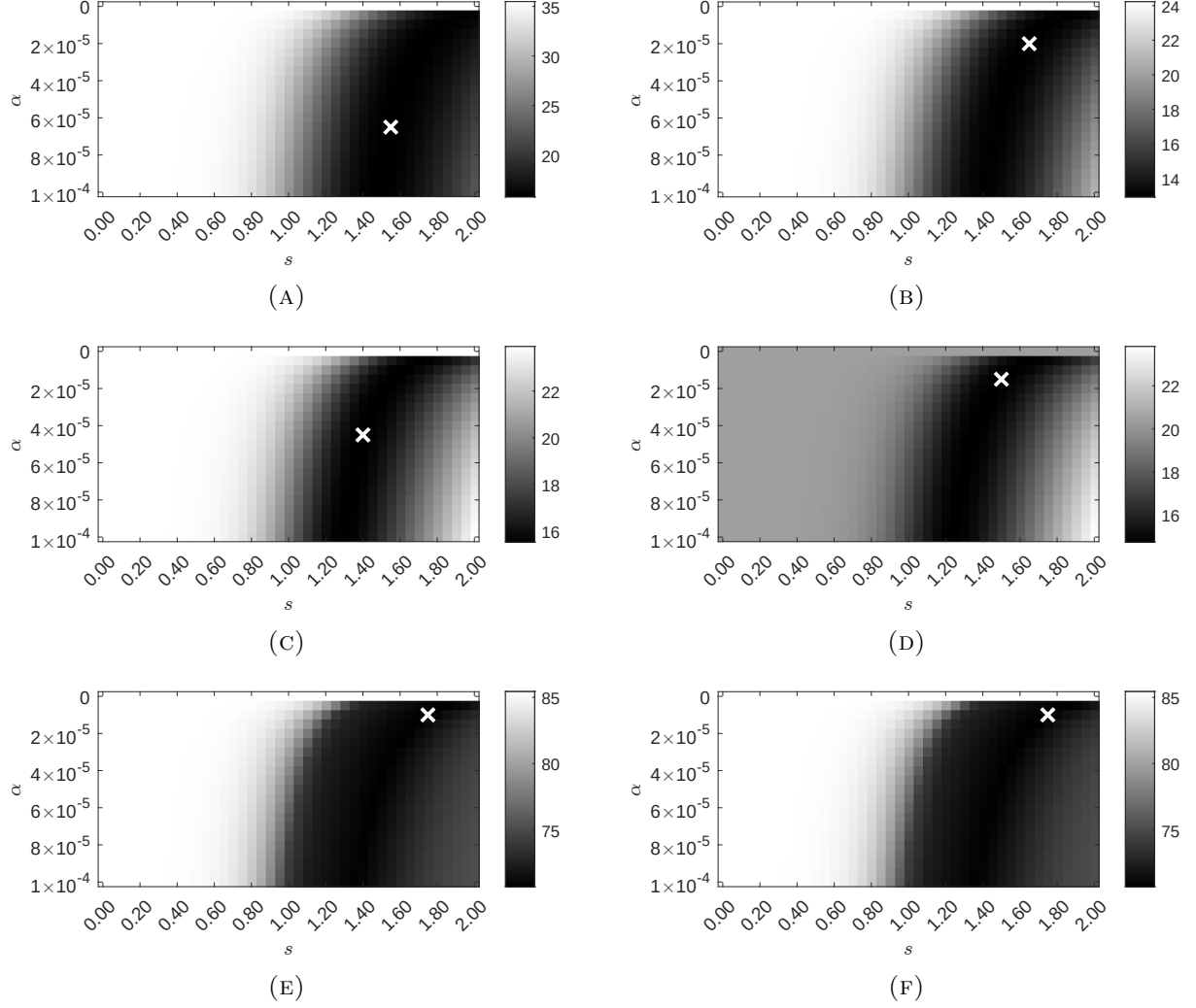


FIGURE S5. Star TCT reconstruction errors as functions of the regularization parameters α and s for the flag phantom, without (left column) and with (right column) the positivity constraint. Rows correspond to the L^1 , L^2 , and L^∞ error metrics.

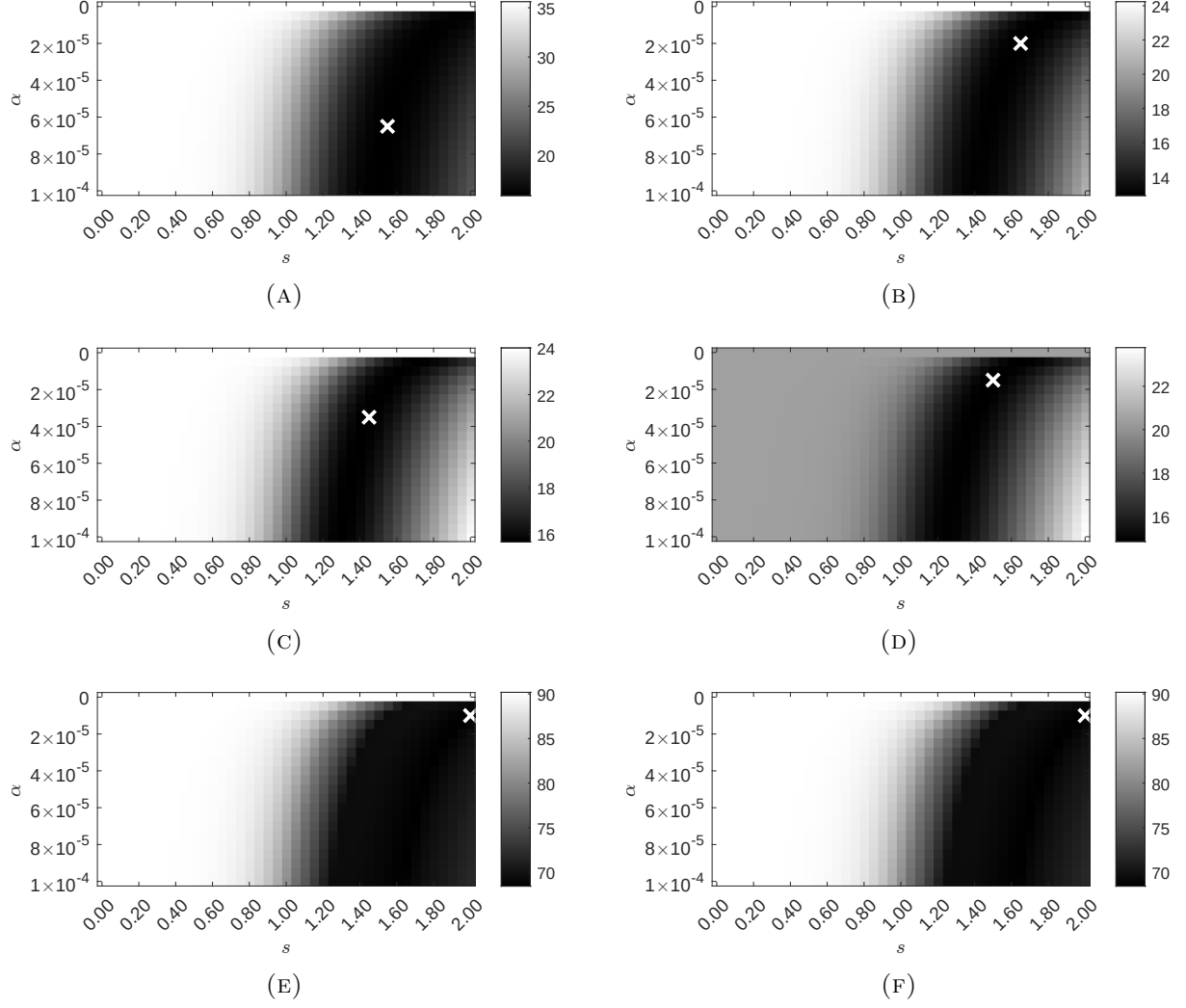


FIGURE S6. Star TCT reconstruction errors as functions of the regularization parameters α and s for the rotated flag phantom, without (left column) and with (right column) the positivity constraint. Rows correspond to the L^1 , L^2 , and L^∞ error metrics.

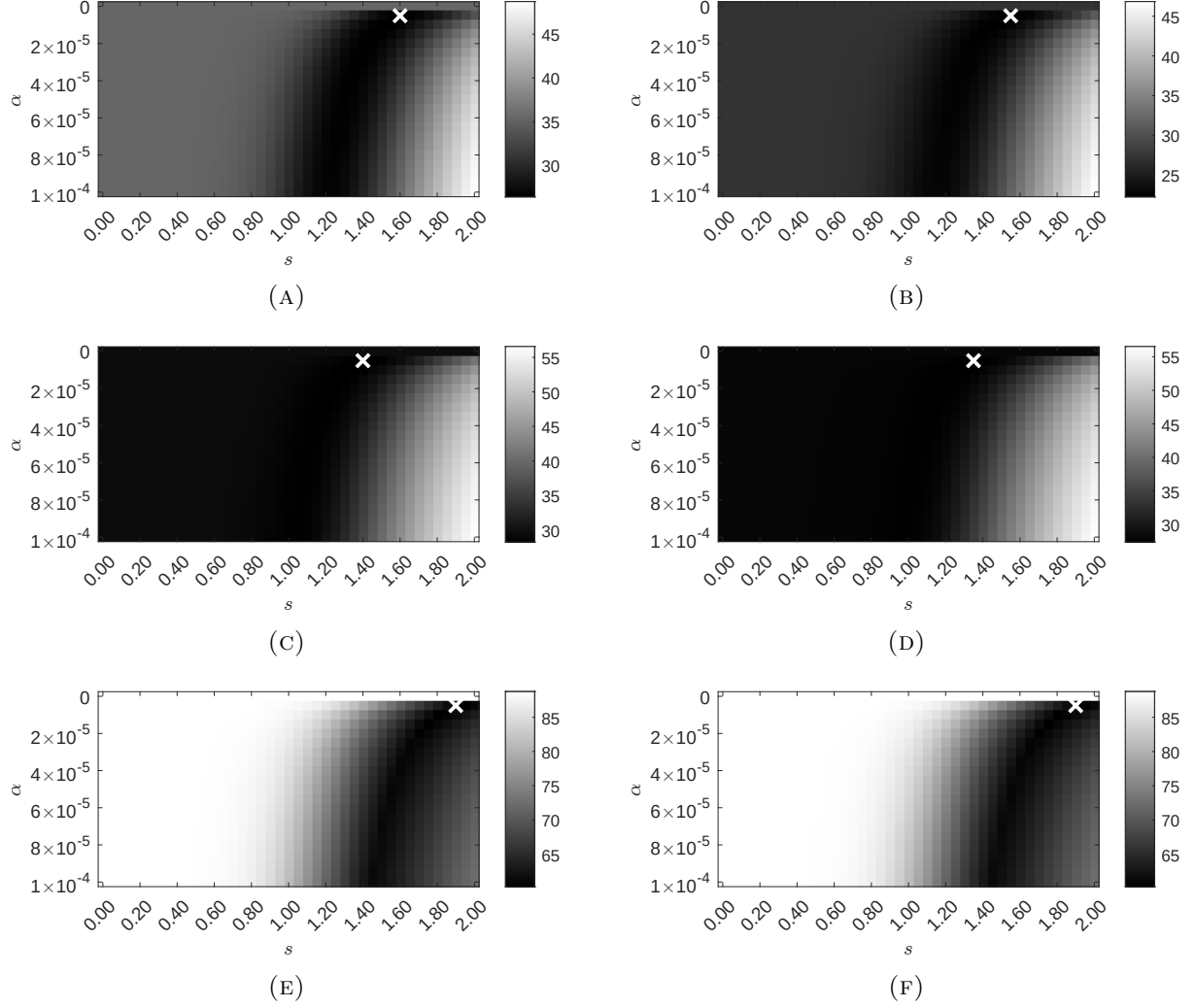


FIGURE S7. f/cFTBP reconstruction errors as functions of the regularization parameters α and s for the Shepp–Logan phantom, without (left column) and with (right column) the positivity constraint. Rows correspond to the L^1 , L^2 , and L^∞ error metrics.

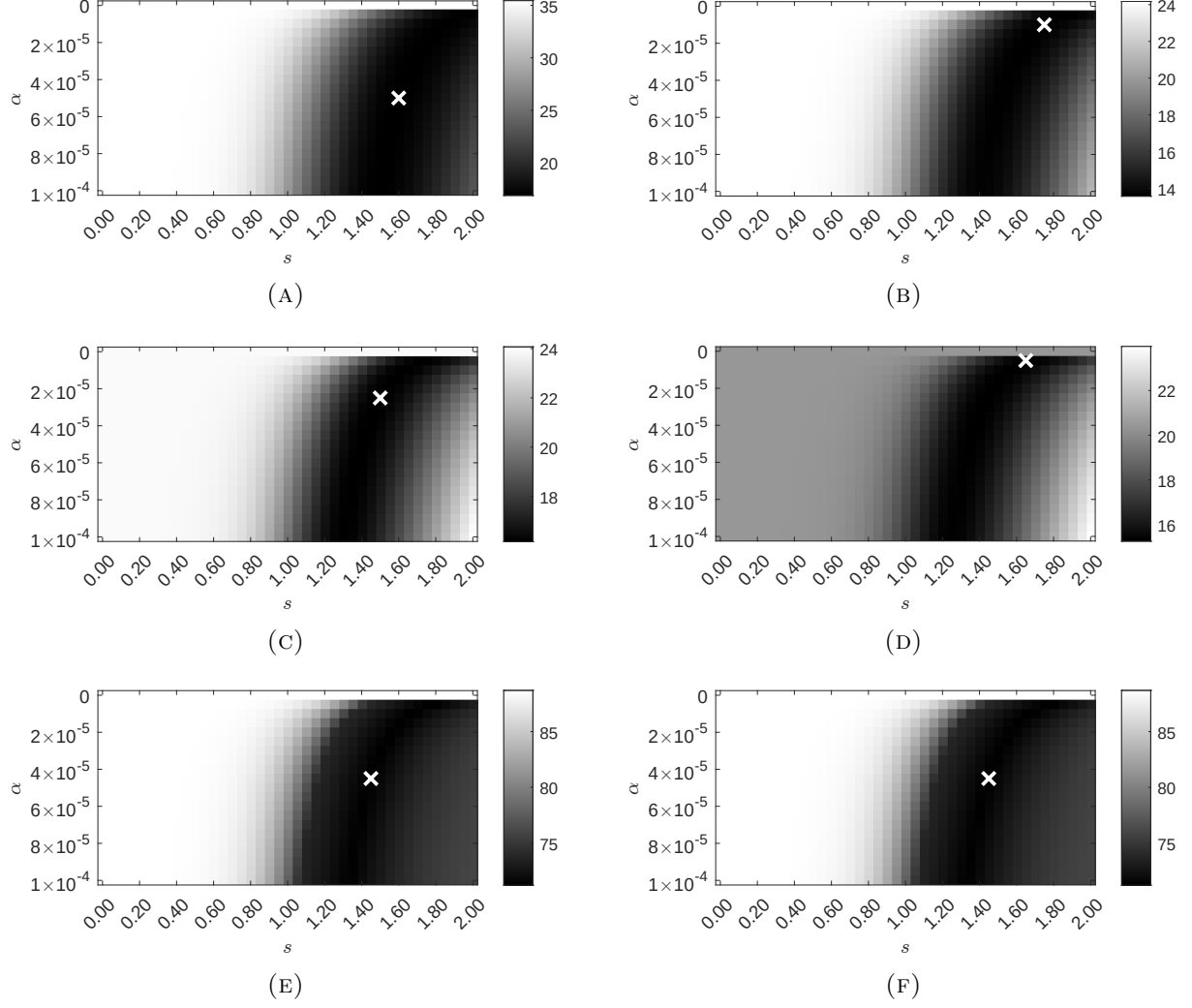


FIGURE S8. f/cFTBP reconstruction errors as functions of the regularization parameters α and s for the flag phantom, without (left column) and with (right column) the positivity constraint. Rows correspond to the L^1 , L^2 , and L^∞ error metrics.

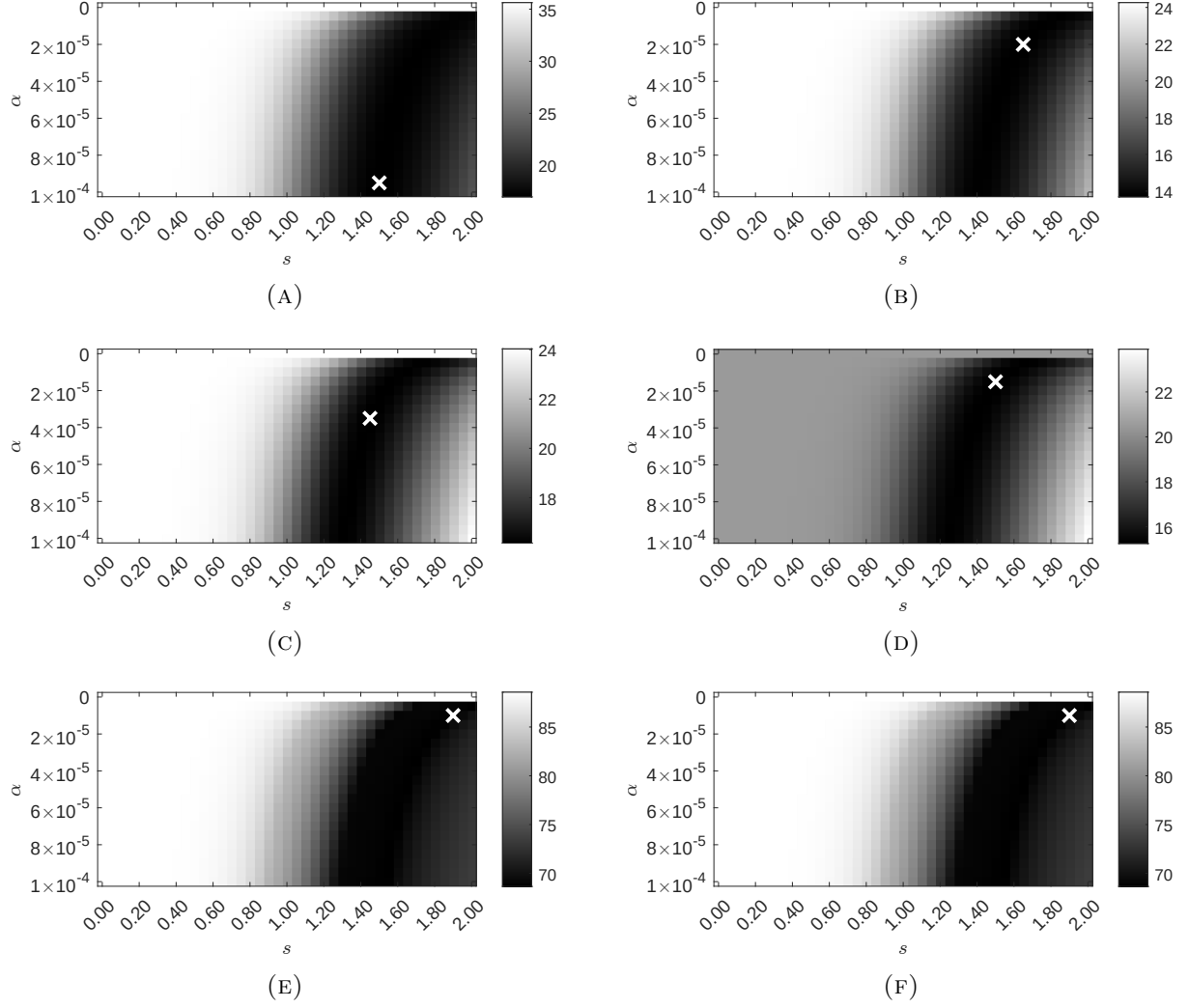


FIGURE S9. f/cFTBP reconstruction errors as functions of the regularization parameters α and s for the rotated flag phantom, without (left column) and with (right column) the positivity constraint. Rows correspond to the L^1 , L^2 , and L^∞ error metrics.